



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai: 400058, India

(An Empowered Autonomous College Affiliated to University of Mumbai)

Department of Computer Engineering

Vehicle Rental Management System – Activity Diagram

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Aim:

To draw an Activity diagram for the Vehicle Rental Management System.

Implementation:

The activity diagram below models the complete behavior of the Vehicle Rental Management System. The flow of control was derived from the use cases of Experiment 2, the classes and operations of Experiment 3 and the interactions of Experiment 4, so that every activity shown corresponds to a documented requirement of the Software Requirements Specification. Nine swimlanes separate the responsibilities of the Guest, the Customer, the Branch Staff, the VRMS System, the Payment Gateway, the Notification Service, the Maintenance Technician, the Administrator and the Branch Manager. Because the system is large, the single activity has been drawn in five consecutive phases; the circled connectors (A) to (D) carry the control flow from one phase to the next.

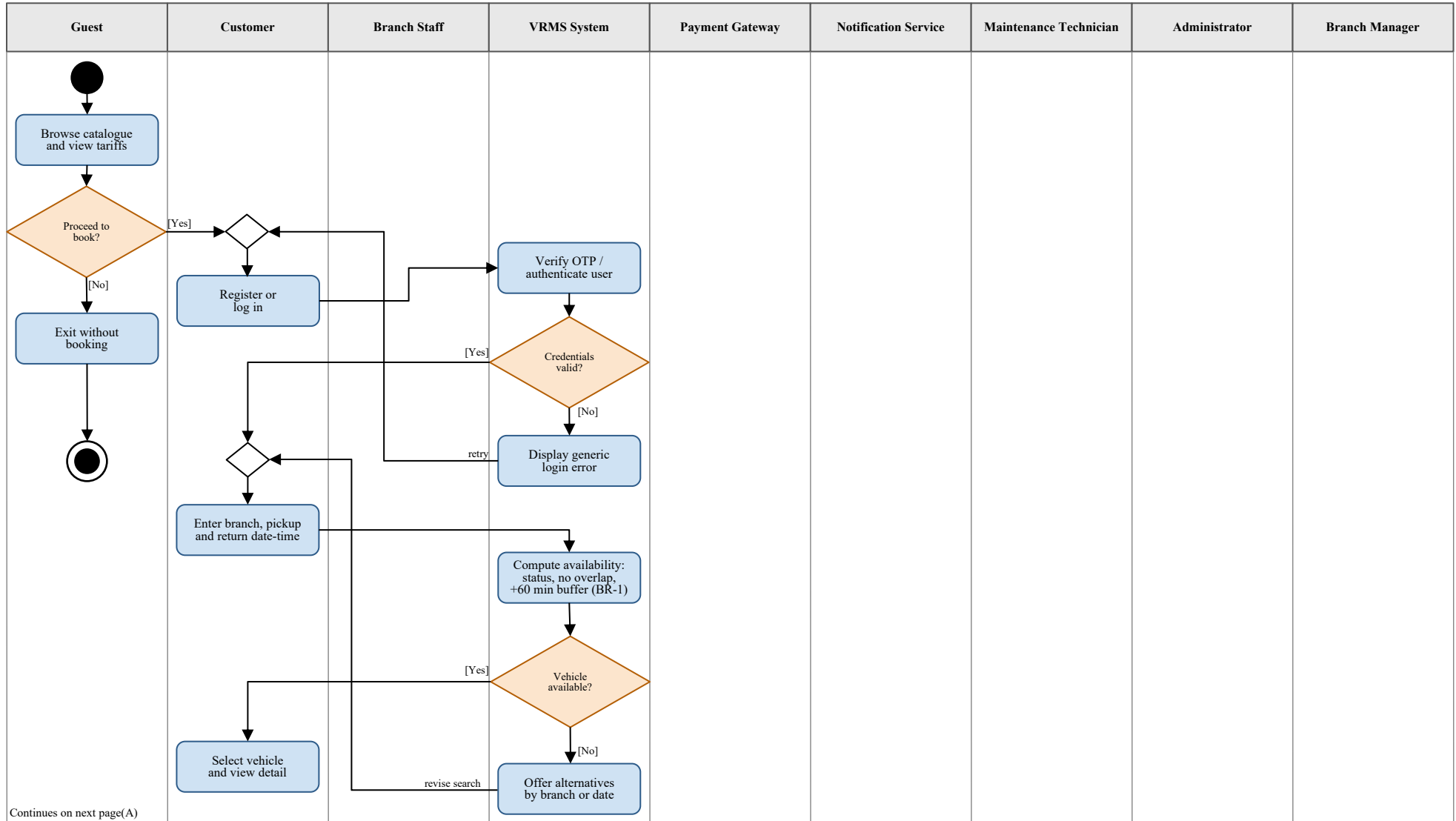
Notation used:

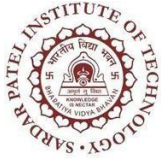
- Initial state – a filled circle marking the start of the activity.
- Final state – a filled circle inside a ring marking the end of the activity.
- Activity state – a rounded rectangle naming an action performed in that swimlane.
- Decision node – a diamond with one incoming and two or more outgoing flows.
- Guard – a condition in square brackets written beside an outgoing flow.
- Merge node – a small diamond where alternative flows come back together.
- Fork – a solid bar splitting one flow into several concurrent flows.
- Join – a solid bar where concurrent flows synchronize into one.
- Time event – an hourglass marking a flow triggered by the passage of time.
- Control flow – a solid arrow showing the order of execution.
- Swimlane – a labelled vertical band grouping the activities of one actor.



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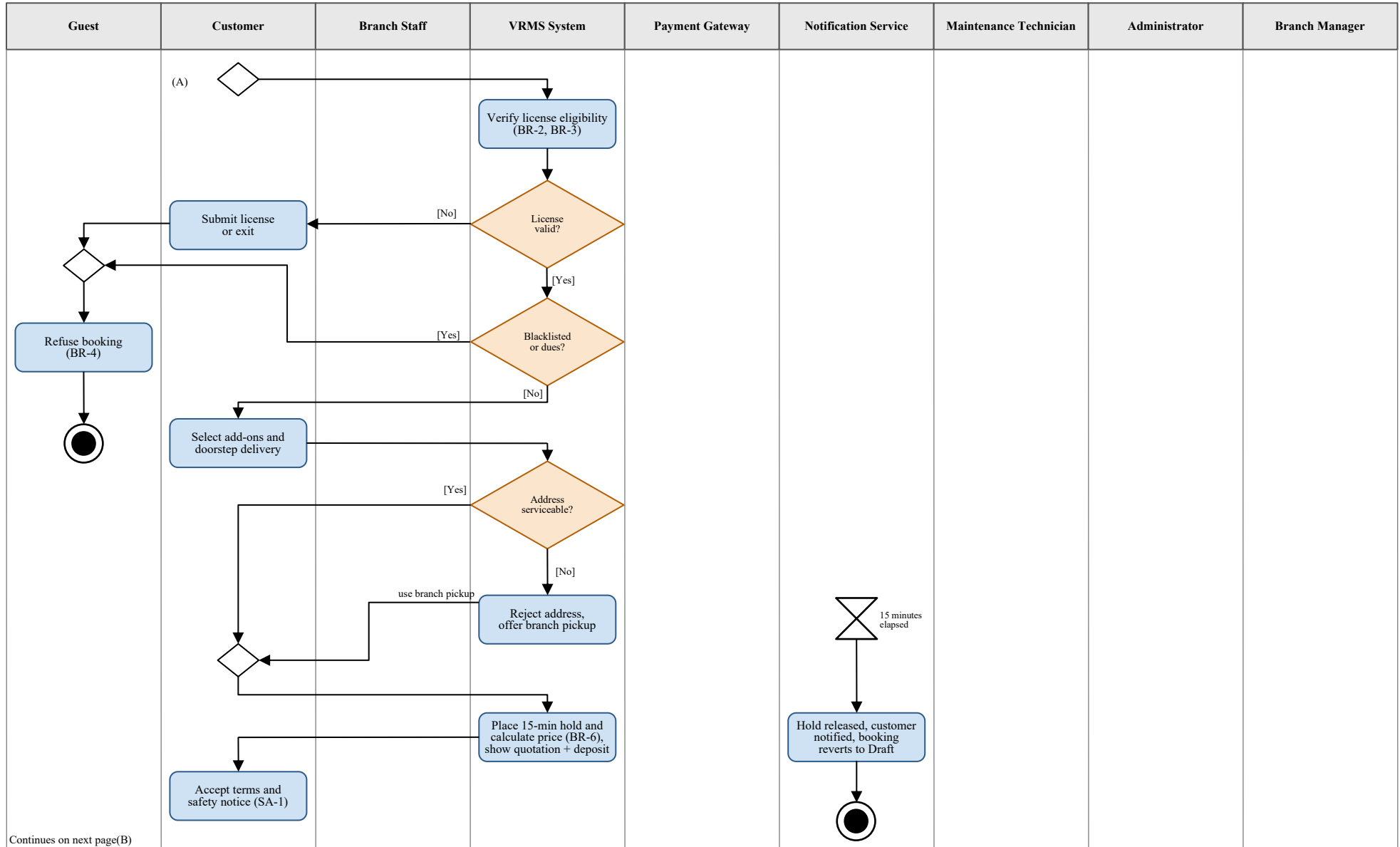
Phase 1 – Access, Registration and Vehicle Search





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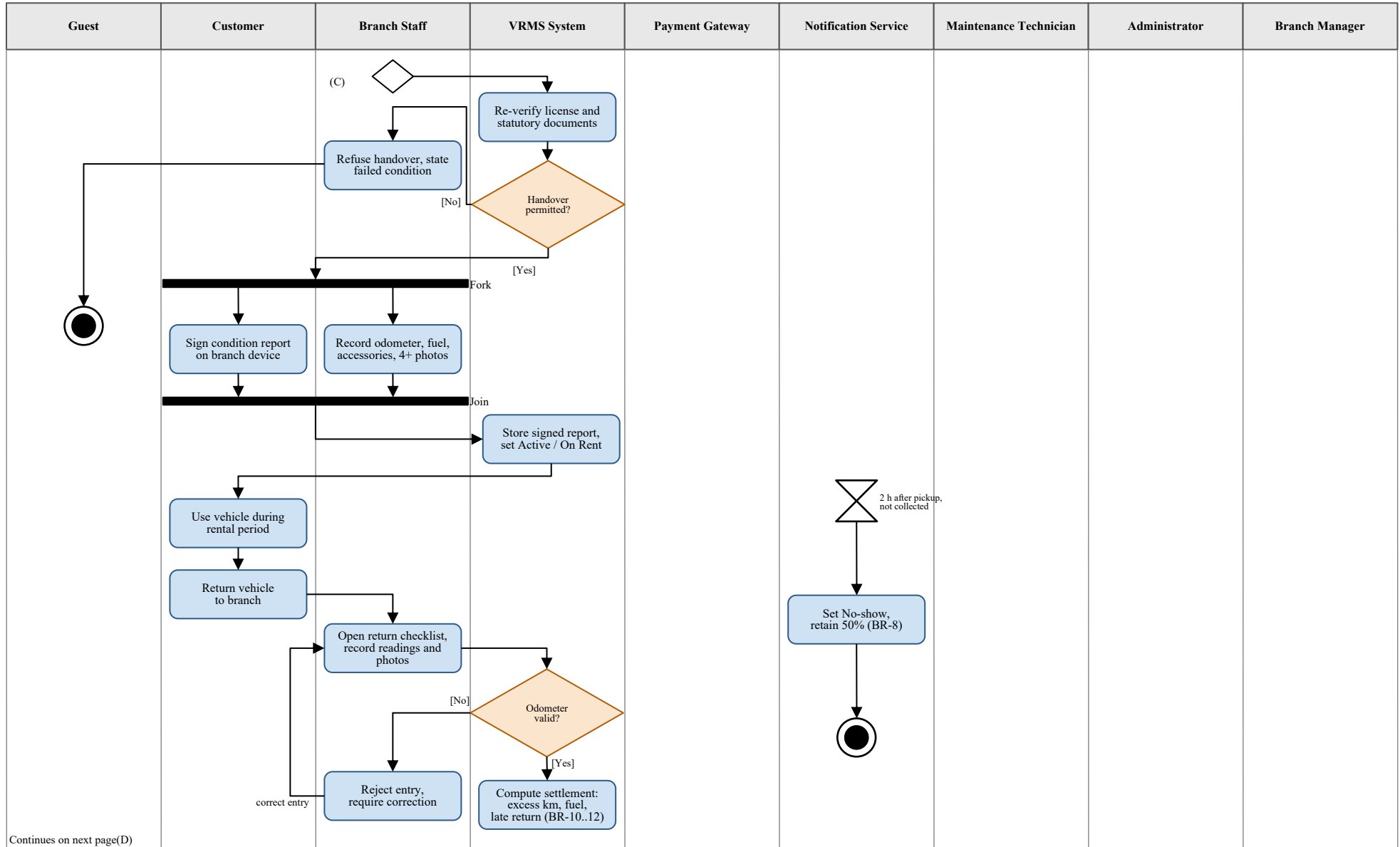
Phase 2 – Eligibility, Add-ons and Quotation





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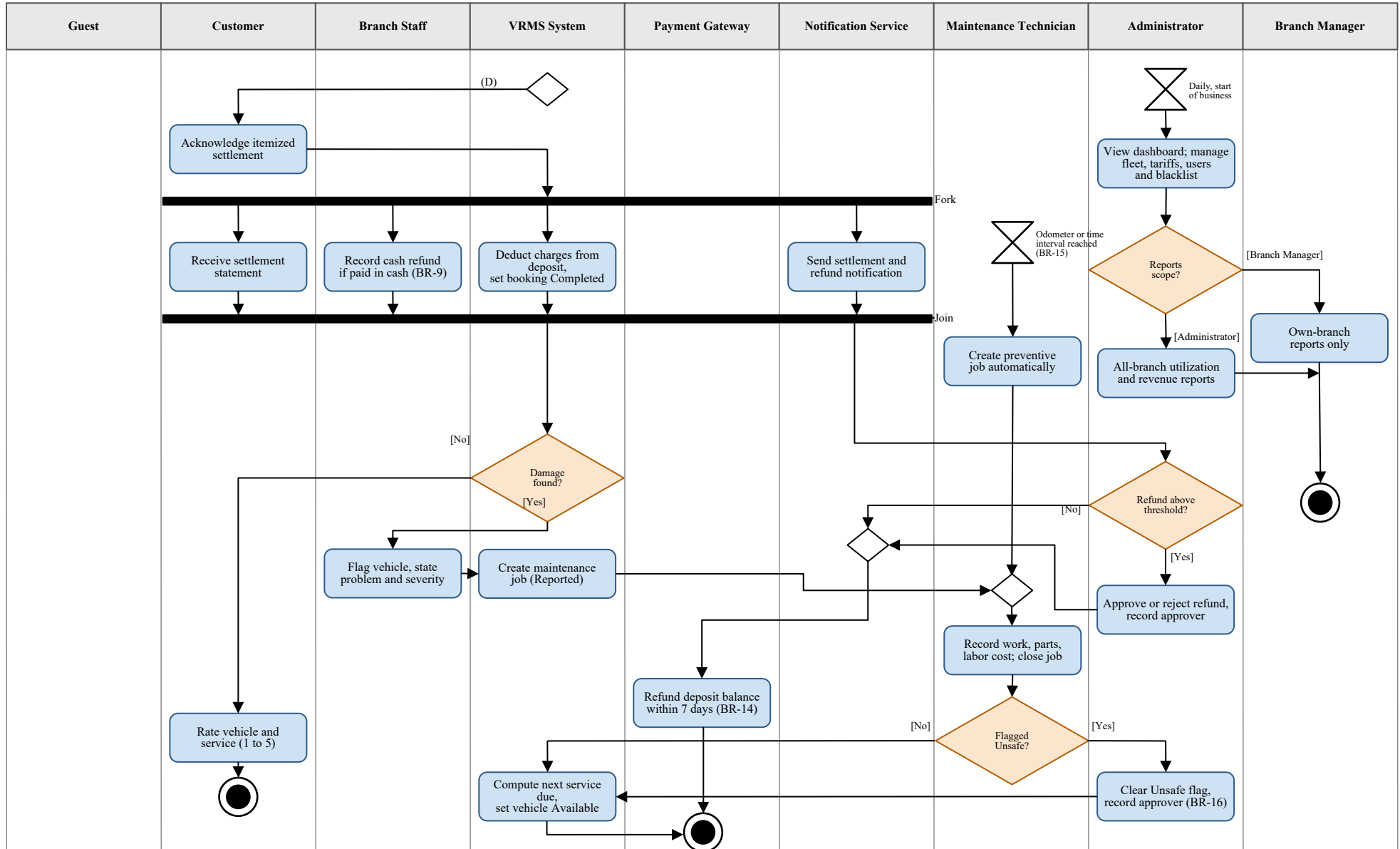
Phase 4 – Rental Period, Return and Settlement

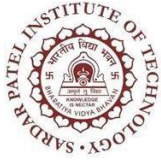




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Phase 5 – Settlement, Maintenance and Administration





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Conclusion:

By performing this experiment, we were able to model the dynamic behavior of the Vehicle Rental Management System using an activity diagram. We learned how to allocate actions to swimlanes so that the responsibility of every actor is explicit, how to branch a flow with decision nodes and guards and bring it back together with merge nodes, how to express concurrent work with fork and join bars, and how to start a flow from the passage of time using time events. The three fork and join pairs model the atomic confirmation of a booking, the simultaneous recording and signing of the condition report at handover, and the parallel settlement carried out when a vehicle is returned. The diagram is consistent with the use cases, the class diagram and the sequence and collaboration diagrams prepared earlier, and every activity is traceable to a requirement or business rule of the Software Requirements Specification.